INCH-POUND
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COMMERCIAL ITEM DESCRIPTION

ANALYZER, DIGITAL DATA

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. This CID defines the minimum technical and performance requirements for a digital data analyzer, also known as "Common Aircraft Portable Reprogramming Equipment" (CAPRE). The equipment will function as a portable flight-line programmer in conjunction with Aircraft Adapter Groups (AAGs) for the reprogramming and downloading of mission data files and operational flight programs during preparation and maintenance of aircraft Line Replaceable Units (LRUs) and support equipment. The analyzer retrieves the encrypted mission critical data from various secured defense networks and uploads to LRUs. This equipment also downloads the recorded mission critical data from LRUs and uploads to various secured defense networks and ground test stations for analysis.

2. SALIENT CHARACTERISTICS.

2.1 System overview. The digital data analyzer consists of a laptop computer with peripheral interfaces, an operating system, and an external device interface, which provides the capability to communicate with an AAG. The analyzer is classified as Class 1 equipment in the latest version of MIL-PRF-28800F "Test Equipment for Use with Electrical and Electronic Equipment, General Specification for."

2.2 Safety. All equipment shall include fail safe features for safety of personnel during installation, operation, maintenance, and repair activities. The analyzer shall meet all safety requirements for equipment outlined in MIL-PRF-28800 paragraph 3.10, as well as the following subparagraph.

2.2.1 Shielding. Protective shields shall be provided to protect personnel from accidental contact with electrical parts during maintenance actions. All exposed metal surfaces of the controller chassis shall be capable of being tied to the aircraft ground. Any power transformers shall be located inside the laptop unit.

2.3 Electrical power sources and connections. The analyzer shall operate from nominal commercial, military, and shipboard power sources operating at AC voltages of 115 - 240 VAC at 50 - 400 Hertz. The analyzer shall also be capable of operating with an internal battery source and from an external source of 10 – 30 VDC. Connectors shall withstand a high number of mate/demate actions. All power cables shall be designed for use in the flight-line environment and allow for organizational level repair.

2.3.1 Interruption of power source. Voltage or frequency interruptions shall not cause damage to the equipment. After power interruptions of any duration, the analyzer shall automatically resume operation at either the last operating condition before interruption, or a default power up condition.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data that may improve this document should be sent to: Warner Robins Air Logistics Center, WR-ALC/TILCC, Robins AFB., GA 31098-1640

FSC 6625

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2.3.2 Internal removable battery. The internal removable battery when fully charged, shall provide a minimum of 90 minutes of continuous operation. The controller shall charge an installed battery when powered from an external source and incorporate smart charging techniques in order to maximize battery life.

2.4 Equipment components. Parts, materials, and processes concerning electronic components, which are part of this equipment, shall be as specified in MIL-PRF-28800 paragraphs 3.2 through 3.2.3.3.

2.5 Performance and functional requirements. The analyzer provided shall meet the following minimum performance and functional requirements. Compliance with these requirements shall be verified in accordance with Section 4.

2.5.1 CAPRE controller. The controller shall be PC compatible and shall be compatible with Windows 2000 and higher.

2.5.2 Central Processing Unit (CPU). The CPU shall be compatible with and equivalent in performance to an Intel Pentium II, 300 MHz with 512kb of L2 cache, or faster.

2.5.3 Memory. The controller shall have a minimum of 128 Megabytes of memory.

2.5.4 Video adapter. The controller's video adapter shall support resolutions of at least 800 by 600 pixels. The video adapter shall contain a minimum of 2 Megabytes of memory.

2.5.5 USB Interface. The controller shall have at least one root hub USB interface that is environmentally protected.

2.5.6 Data display screen. The color digital display screen shall be have a minimum 10.4 inches diagonal of viewable screen area. The text displayed on screen shall be readable in day, night and sunlight at ±45 degree from screen surface. The screen shall be able to be tilted between 0 - 135 degrees and shall be capable of remaining at the desired angle during normal use.

2.5.7 Input devices. The controller shall include an embedded keyboard with a minimum of U.S. Standard 85 style keys and an embedded pointing device. The keyboard and pointing device shall be designed in a manner which affords protection from rain and dust. The operator shall be able to successfully operate this equipment while wearing bio-chemical protection gear (see 2.6.11).

2.5.8 Removable drives. The controller shall be capable of housing a minimum of two removable drives that shall be operated and are addressable by the operating system simultaneously. The first housing shall be capable of supporting one hard drive. Additional housings shall provide the capability of supporting a CD-ROM or DVD-ROM, a hard drive, and a 3.5 inch, 1.44 Mb floppy drive (not simultaneously).

2.5.8.1 Hard drives. The controller shall be delivered with two hard drives. Each hard drive's formatted capacity shall be at least 2.1 Gigabytes without using drive compression software. Solid state hard drives are acceptable as long as the installed operating system and BIOS recognize and communicate with the drive as it would a magnetic rotating drive. Each hard drive shall be preloaded with Windows Professional 2000, or later version, fully configured to operate the controller. Each hard drive shall also have the latest version of Microsoft Office Professional Edition pre loaded. Each hard drive shall come with Fortezza and an HTML 3.0 compatible Internet browser preinstalled.

2.5.8.2 CD-ROM drive. The controller shall be delivered with a CD-ROM drive. The CD-ROM drive shall be

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capable of operating at a minimum of 8X standard speed. If the controller has an internal non-removable CD-ROM drive, the second housing need not support a CD-ROM.

2.5.8.3 Floppy drive. The controller shall be delivered with a 3.5 inch, 1.44 Megabyte floppy drive. If the controller has an internal non-removable 3.5 inch, 1.44 Mb floppy drive, the second housing need not support a 3.5 inch, 1.44 Mb floppy drive.

2.5.9 Peripheral interfaces. The controller shall have the following functional standard interfaces available for use *in a controlled environment. When peripheral interfaces are not in use, the controller shall provide a means to* protect these interfaces from rain and dust:

a. EPP/ECP bi-directional parallel port

b. Keyboard connector

c. Mouse connector

d. External monitor connector

e. EIA RS-232, 9-pin port connector

f. Two PCMCIA Type 2 slots (unoccupied)

2.6 Environmental requirements. The CAPRE shall meet all of the environmental requirements in the following subparagraphs.

2.6.1 Dimensions and weight. The laptop unit shall be such as to allow one person to easily pick it up, carry it, and raise it over one's head. The controller shall not weigh more than 14 pounds.

2.6.2 Temperature, non-operating. The equipment shall conform to the specified performance and accuracy after being stored in the temperature range of -20° to +71°C. Testing shall be in accordance with MIL-PRF-28800 paragraphs 3.8.2.1.2 and 4.5.5.1.

2.6.3 Temperature operating. The equipment shall conform to the specified performance and accuracy while being operated in a temperature range of –20° to +50°C. Testing shall be in accordance with MIL-PRF-28800 paragraphs 3.8.2.2 and 4.5.5.1.

2.6.4 Relative humidity. The analyzer shall conform to the specified performance and accuracy while exposed to environments with 5 - 85 percent relative humidity in the temperature range of 30 to 40°C in accordance with MIL-PRF-28800 paragraphs 3.8.2.3 and 4.5.5.1.

2.6.5 Random vibration. The analyzer shall conform to the specified performance and accuracy after being subjected to the random vibration levels of 5 - 500 Hz. Testing shall be in accordance with MIL-PRF-28800 paragraphs 3.8.4.1 and 4.5.5.3.1.

2.6.6 Shock. The analyzer shall meet its performance and accuracy requirements in an operating environment after being subjected to a 30g half-sine shock test as outlined in MIL-PRF-28800 paragraphs 3.8.5.1 and 4.5.5.4.1.

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2.6.7 Bench handling. This equipment shall be able to withstand normal shipping and handling occurring in the work area. Testing shall be in accordance with MIL-PRF-28800 paragraphs 3.8.5.3 and 4.5.5.4.3.

2.6.8 Splash-proof. The CAPRE controller, all installed devices, both permanent and removable, and cables shall be encased in accordance with MIL-STD-108. The equipment shall conform to specified performance after being tested in accordance with the splash-proof test defined in MIL-PRF-28800 paragraph 3.8.6.2.

2.6.9 Explosive atmosphere. The CAPRE controller and all installed devices, both permanent and removable, shall be fully functional in an explosive atmosphere and after being subjected to the explosive atmosphere tests outlined in MIL-PRF-28800 paragraph 4.5.6.3.

2.6.10 Dust resistance. The CAPRE system, when not in a transit case, shall withstand the blowing dust test described by MIL-STD-810, Method 510.3. Penetration of dust particles is allowable as long as the analyzer can be returned to a fully functional and operational status after being cleaned. The CAPRE system shall be capable of being disassembled, cleaned and reassembled in one hour or less when dust needs to be removed.

2.6.11 Ease of use in a contaminated environment. The CAPRE shall be designed to be used by an operator wearing bio-chemical warfare gear.

2.6.12 Exposure to fluids requirement. Exposed surfaces shall withstand, without damage, any fluids commonly encountered in operational and maintenance areas, such as aviation fuel, hydraulic fluid, and various types of coolant fluid.

2.6.13 Fungus resistance. The equipment shall not contain materials that provide nutrients for the growth of fungus. When a transit case is provided, it shall not contain materials that provide nutrients for the growth of fungus.

2.6.14 Solar radiation. The equipment shall not be damaged and shall maintain specified performance while exposed to sunlight (solar radiation) as specified in MIL-PRF-28800 paragraph 4.5.6.8.

2.7 Electromagnetic Interference (EMI). The analyzer shall operate in the intended environment unaffected by Electromagnetic Interference (EMI) as stated in MIL-PRF-28800F, paragraphs 3.9 and 4.5.6.5.

2.8 Electrostatic Discharge (ESD) control. This equipment shall comply with performance control requirements concerning electrostatic discharge sensitivity (ESD) as outlined in MIL-STD-1686C.

2.9 Panel markings, warning label, and product identification:

2.9.1 Panel markings. Each analyzer shall be provided with a panel marking, which enables the operator to identify the functions and use of all items requiring operator intervention.

2.9.2 Warning label. A warning label shall provide information about the possible location, nature and the extent of any hazard. As stated in MIL-PRF-28800 paragraph 3.11, warning markings shall have a high contrast between the letter and background colors. The label markings shall be as permanent as the normal life expectancy of the equipment and located as close as possible to the point of danger. Warning markings shall be in accordance with the requirements outlined in ANSI/ISA-S82.01-1994 or ANSI/ISA-S82.02-1994, or both as applicable.

2.9.3 Identification plates. Two identification plates may be required. One shall be affixed to the analyzer in a readable position; the second identification plate shall be affixed to the transit case if one is provided. Each shall contain the following information in legible and permanent print:

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a. Equipment name:

b. Manufacturer's model and/or serial number:

c. National Stock Number (NSN):

d. Contract Number:

2.10 Reliability tests. Reliability verification shall conform with MIL-PRF-28800 paragraphs 3.13 and 4.5.8.1. The equipment shall operate for 2,000 hours without failure with a statistical certainty of 95 percent.

2.11 Maintainability. Maintenance provisions shall be completed in accordance with MIL-PRF-28800 paragraphs 3.14.1, 3.14.2, 3.14.3, and 4.5.7.1.1. The manufacturer shall provide all maintenance manuals, procedures, equipment, instructions, drawings, electrical schematics, etc. required for maintenance, troubleshooting, and repair.

2.12 Y2K compliance. The analyzer shall be Y2K Year 2000 compliant and shall accurately process date ranges and time data from, into, and between the 20th and 21st Centuries; specifically, the years 1999, 2000, 2001, and leap year calculations. This compliance includes comparison and sequential date/time data. The analyzer shall also accurately process date and/or time data if such data is received or exchanged from other equipment or sources.

2.13 Features:

2.13.1 Software expandability. The capability for software modifications and upgrades shall be provided. Software modifications or upgrades shall not require opening or disassembly of the analyzer.

2.13.2 Hardware expandability. The analyzer shall include provisions for hardware upgrades. Expansion slots, removable/replaceable components, expandable memory, or other means may be used.

3. REGULATORY PROVISIONS.

3.1 Recoverable materials. In accordance with Section 23.403 of the Federal Acquisition Regulation (FAR), the Federal Government's policy is to acquire items composed of the highest percentage of recoverable materials practicable consistent with maintaining a satisfactory level of competition and without adversely affecting performance requirements or exposing the supplier's employees to undue hazards from the recovered materials.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Product conformance. The contractor shall certify and maintain substantiating evidence that the digital data analyzer offered meets the salient characteristics of this CID, conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same product offered for sale in the commercial market. The Government requires proof of such conformance prior to first delivery and thereafter, as may be otherwise provided under the provisions of the contract. Information substantiating product compliance with the requirements of this CID, paragraph by paragraph, shall be submitted concurrently with the bid sample unless otherwise specified in the contract.

4.2 Bid sample(s). The offeror shall submit a bid sample for Government test and evaluation in accordance with the solicitation. Government evaluation will consist of inspections, demonstrations, and tests to assess compliance with the requirements of this CID and to determine whether the digital data analyzer provides accurate performance and has acceptable form, fit, function and suitability for use. The bid sample shall be submitted concurrently with the offeror's proposal to the address specified in the solicitation. The bid sample shall be submitted at no cost to

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Government expense and returned to the sender at the sender's request, upon completion of the test.

5. PACKAGING.

5.1 Packaging. Preservation, packing, and packaging requirements for this analyzer shall be as specified in the contract.

6. NOTES.

6.1 Document ordering.

a. Copies of FARs and Department of Defense specifications, standards, and handbooks are available from: Defense Automated Printing Service, (DAPS), Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094. Phone 215/697-2179.

b. Copies of ANSI documents are available from: ANSI, 11 West 42nd Street, New York, NY 10036, Phone 212/642-4900.

6.2 Ordering data. The following information should be provided at the time of contract award:

a. Title, CID document number and date.

b. Bid samples required. (see 4.2).

c. Packaging requirements (see 5.1).

d. Performance requirements (see 2.2 through 2.13.2).

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MILITARY INTEREST:

Custodian:
Air Force – 99
DLA – CC

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA – FSS

Preparing Activity:
Air Force – 84

Agent:
Air Force – 99

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